```
            1 0 1 0 1 0 1
1 0 0 1 ) 1 0 1 1 1 0 0 0
          1 0 0 1
            1 0 1
            0 0 0
            1 0 1 0
            1 0 0 1
                1 1 0
                0 0 0
                1 1 0 0
                1 0 0 1
                0 1 0 1
                  0 0 0
                  1 0 1 0
                  1 0 0 1
                    0 1 1
                      R
```

Check for error

```
            1 0 1 0 1 0 1
1 0 0 1 ) 1 0 1 1 1 0 0 1 1
          1 0 0 1
            1 0 1
            0 0 0
            1 0 1 0
            1 0 0 1
                1 1 0
                0 0 0
                1 1 0 1
                1 0 0 1
                0 1 0 0
                  0 0 0
                  1 0 0 1
```

$$\begin{array}{r} 1\ 0\ 0\ 1 \\ \hline 0 \end{array}$$

No error

P1. Suppose the information content of a packet is the bit pattern 1110 0110 1001 0101 and an even parity scheme is being used. What would the value of the field containing the parity bits be for the case of a two-dimensional parity scheme? Your answer should be such that a minimum-length checksum field is used.

$$\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ \hline 0 & 1 & 0 & 0 & 1 \end{array}$$

P2. Show (give an example other than the one in Figure 6.5) that two-dimensional parity checks can correct and detect a single bit error. Show (give an example of) a double-bit error that can be detected but not corrected.

$$\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ \hline 0 & 1 & 1 & 0 & 1 \end{array} \qquad \begin{array}{cccc|c} 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ \hline 0 & 0 & 1 & 0 & 1 \end{array}$$

- knows which column has error but not which row, as row parity is same.

P5. Consider the 5-bit generator, G = 10011, and suppose that D has the value 1010101010. What is the value of R?

$$\begin{array}{r} 1\,0\,1\,1\,0\,1\,1\,1 \\ 10011\ \big)\overline{1\,0\,1\,0\,1\,0\,1\,0\,1\,0\,0\,0\,0\,0} \\ 1\,0\,0\,1\,1 \end{array}$$

1 1 0 0
0 0 0 0

1 1 0 0 1
1 0 0 1 1

1 0 1 0 0
1 0 0 1 1

1 1 1 1
0 0 0 0

1 1 1 1 0
1 0 0 1 1

1 1 0 1 0
1 0 0 1 1
1 0 0 1 0
1 0 0 1 1

0 1 0
0 0 0

$\underbrace{0\ 1\ 0\ 0}_{R}$

P1:

1	1	1	0	1
0	1	1	0	0
1	0	0	1	0
0	1	0	1	0
0	1	0	0	1

P2:

1	1	1	0	1
0	0	1	0	1
1	0	0	1	0
0	1	0	1	0
0	0	0	0	1

1	1	1	0	1
0	0	1	0	1
1	1	0	1	1
0	1	0	1	0
0	1	0	0	1

P3:

(a) 10011) 10001001010000 = 100100100

10001001010000
10011
―――――
100
000
―――――
1000
0000
―――――
10001
10011
―――――
100
000
―――――
1001
0000
―――――
10010

10011

10
00

100
000

1000
0000

0000

(b)

```
            0101010101
10011 ) 01011010100000
        00000
         10110
         10011
          1011
          0000
          10110
          10011
           1011
           0000
           10110
           10011
            1010
            0000
            10100
            10011
             1110
             0000
```

11100
10011

1111

(c)

0110001010

10011) 01101000110000
00000

11010
10011

10010
10011

10
00

101
000

1011
0000

10110
10011

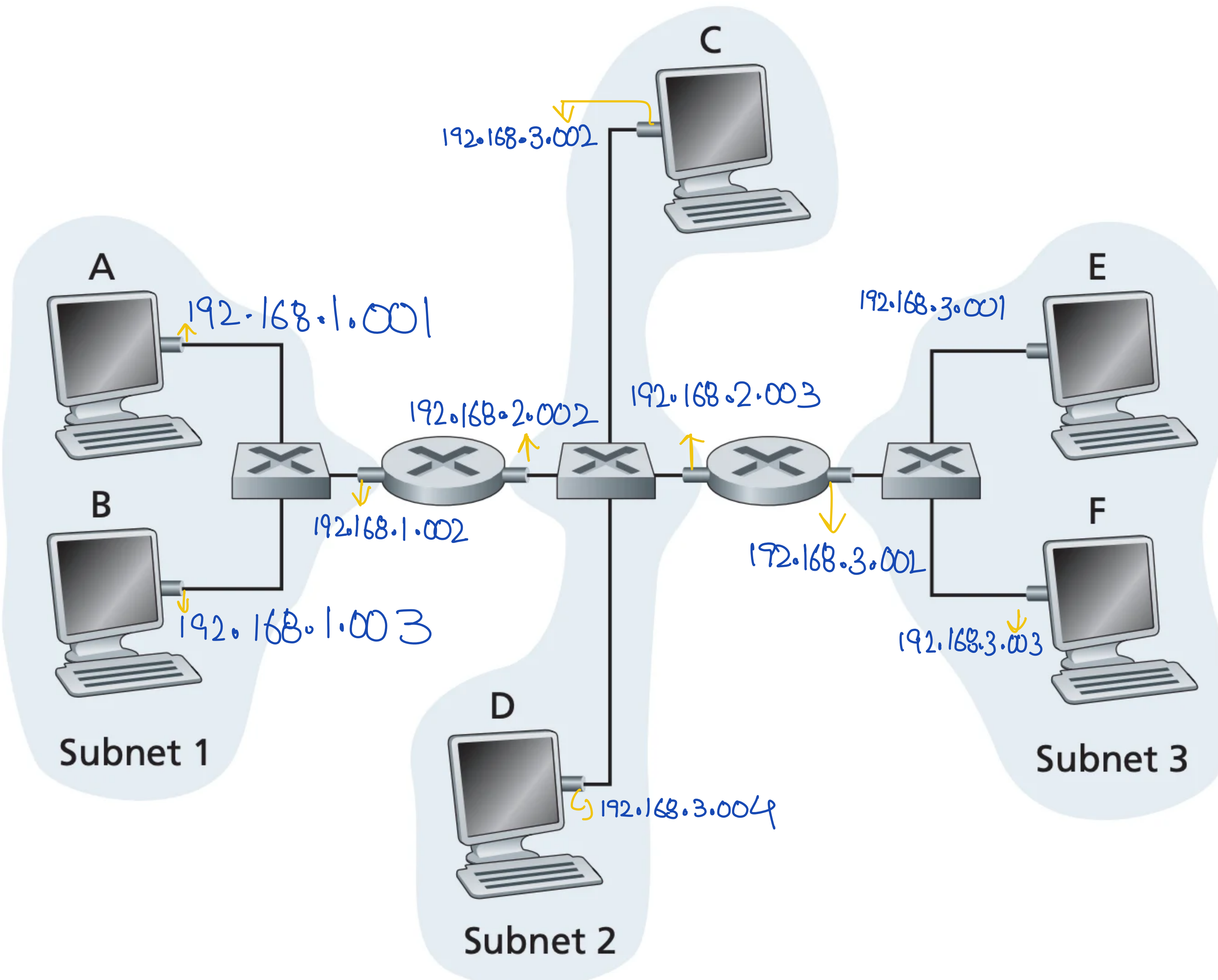
1010
0000

10100
10011

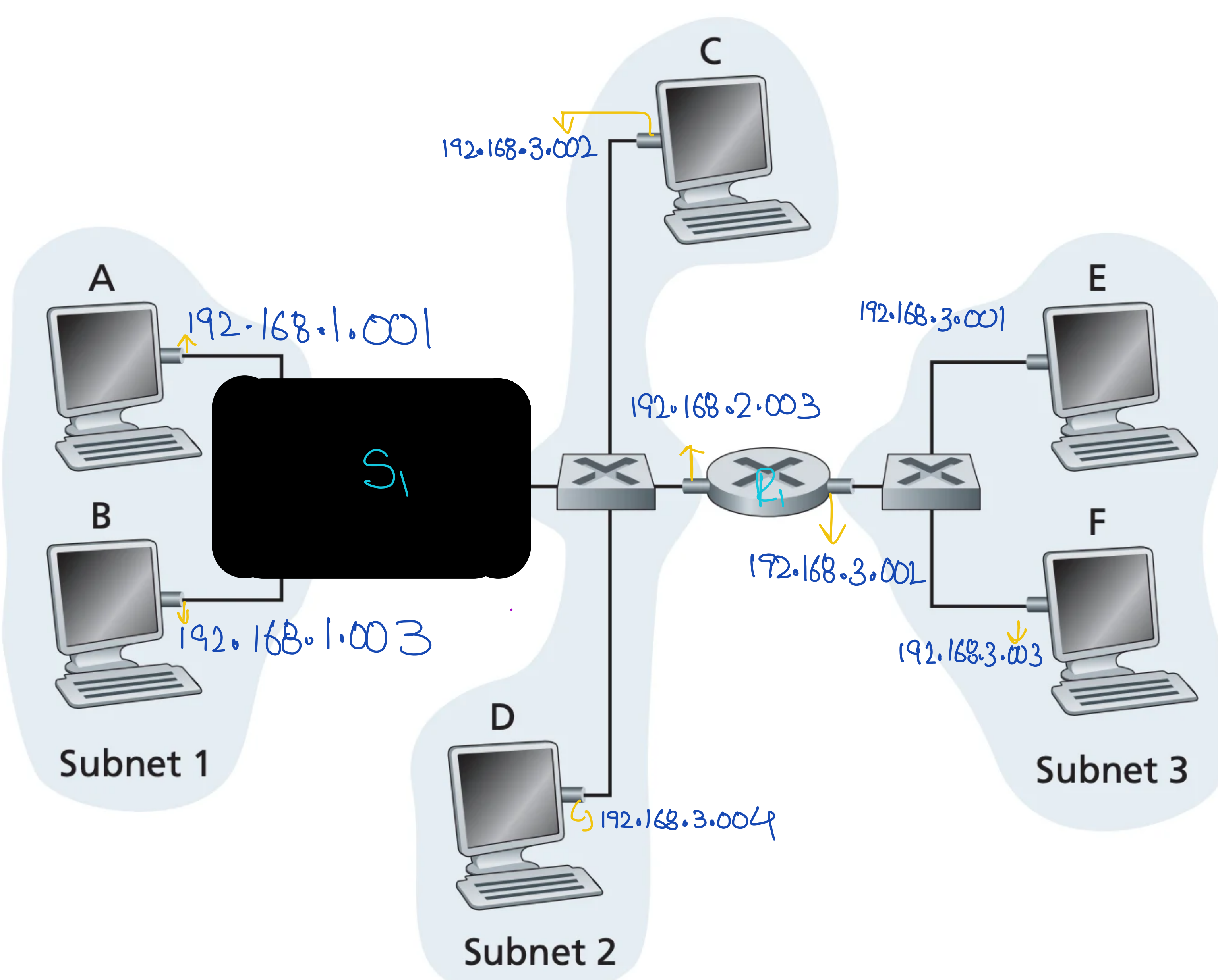
1110
0000

1110

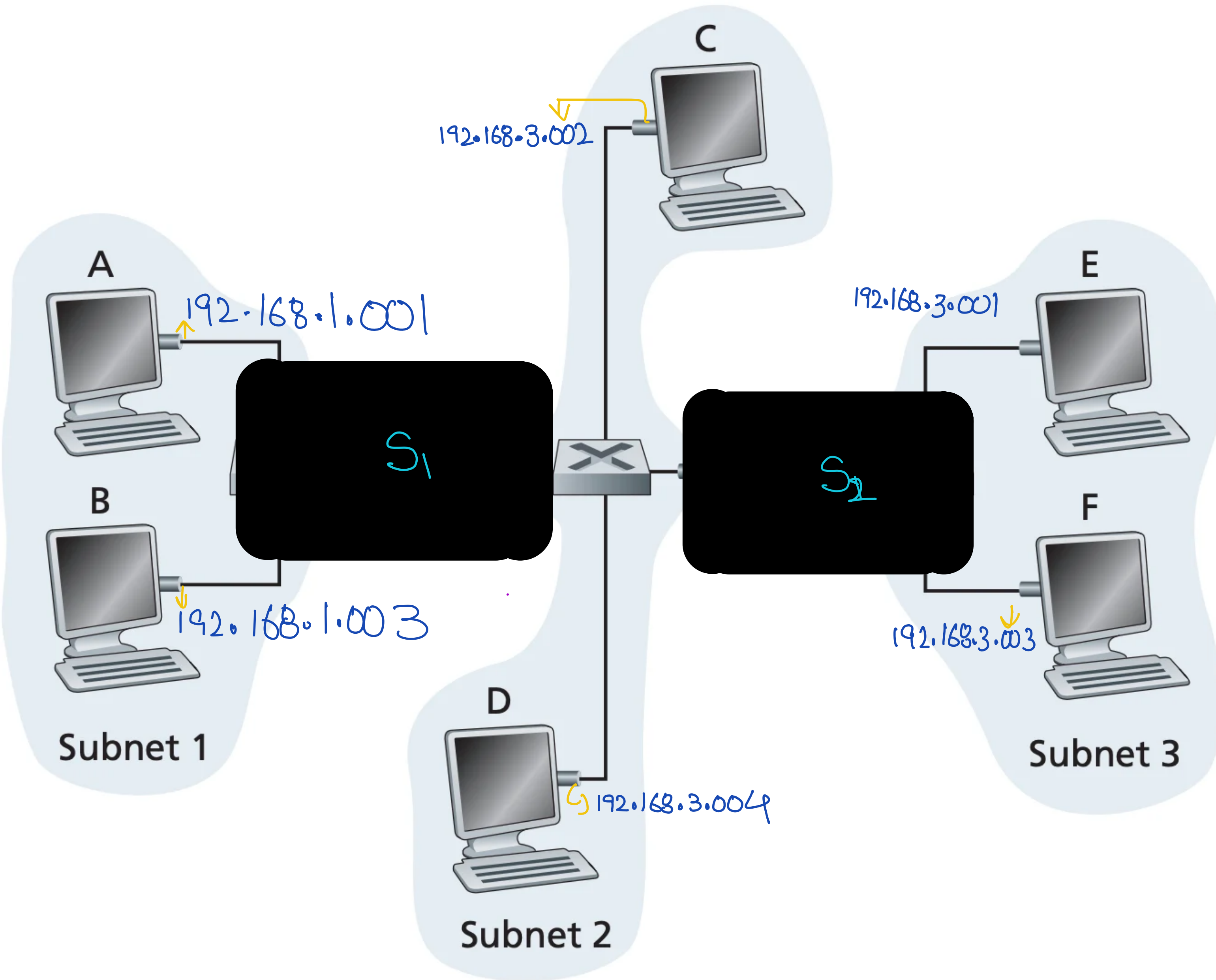
P14


C
192.168.3.002
A
192.168.1.001
192.168.2.002
192.168.2.003
E
192.168.3.001
B
192.168.1.002
192.168.3.002
F
192.168.1.003
192.168.3.003
D
Subnet 1
192.168.3.004
Subnet 3
Subnet 2

P15


C
192.168.3.002
A
192.168.1.001
E
192.168.3.001
192.168.2.003
S_1
R_1
B
192.168.3.002
F
192.168.1.003
192.168.3.003
D
Subnet 1
Subnet 3
192.168.3.004
Subnet 2

P16

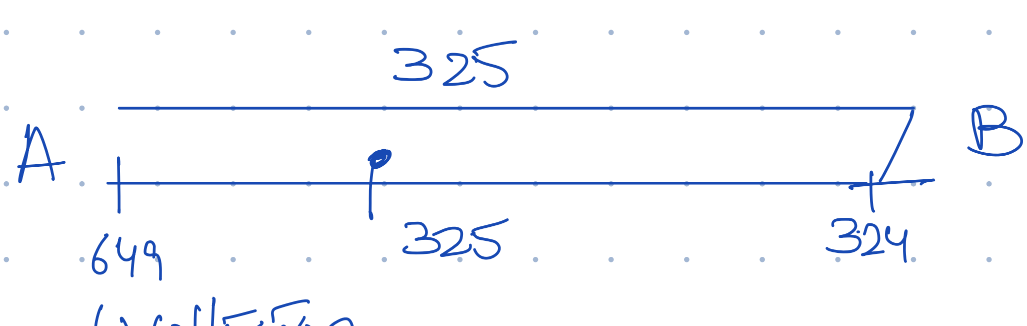


P17.

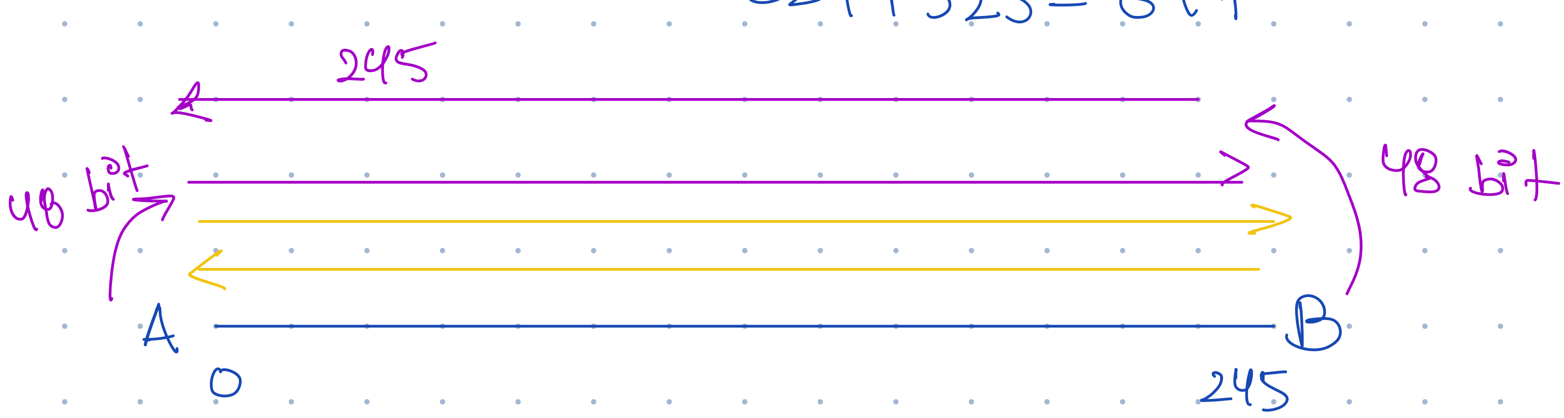
$$100 \times 512 \text{ bits} \times \frac{1 \text{ sec}}{1 \times 2^{30} \text{ bits}}$$

$t = 245$

$t = 512 + 245$



$324 + 325 = 649$



$t=0$ A & B send
$t=245$ A & B detect collision
$t=245+48$ A & B send Jam signal → $293+512=805$

A waits 0 times & B waits 512 times

$t=293+245$ A & B's Jam signal reach each other

$t=538+96$ A starts transmitting

$t=634+245$ A's transmission reaches B

$t=879+96$ B's start transmission

Slotted Aloha Efficiency $N*p*(1-p)^{N-1}$
Pure Aloha efficiency, $N*p*(1-p)^{2(N-1)}$

11101101 10101010	0
01001111 11001000	0
10010101 10011110	1
10010100 11000111	0
10011110 01111110	1
00111101 01000101	0

00110101 00010011 1
11111010 11001010 0
~~01111110 00000110 1~~ 0
11101010 10010000 1

11101010 10010000 1

00110010 00111110 0

01101001 01110011 1

```
        10001010
1001 ) 10011011000
       1001
          1
          0
          10
          00
          10
          000
          1011
          1001
            100
            000
            1000
            1001
               10
               00
               10
```

1: 0.4 ✓ [0.8 – 1.8] ✓ [0.4 – 1.4]
2: 1.2 x
3: 2.2 ✓ [2.6 – 3.6]
4: 2.4 ✓ [2.8 – 3.8]
5: 3.6 x
6: 4.1 ✓ [4.5 – 5.5]
7: 4.3 ✓ [4.8 – 5.8]

8: 4.4 ✓

1: 0.4 ✓ [0.8 — 1.8]
2: 1.2 x
3: 2.2 ✓ [2.6 — 3.6]
4: 2.4 ✓
5: 3.6 ✓ [4.0 — 5.4]
6: 4.1 x
7: 4.3 x
8: 4.4 x

3 → abort

[2.4 — 2.8]

1: 0.7 ✓ [1.1 — 2.1]
2: 1.7 x
3: 2.1 ✓ [2.5 — 3.5]
4: 3.5 ✓ [3.9 — 4.9]
5: 3.8 ✓ [4.2 — 5.2]
6: 4.3 x

$3.8 + 0.4 = 4.2$

1: 0.5 ✓ [0.9 — 1.9]
2: 0.8 ✓ [1.2 — 2.2]
3: 1.2 x
4: 2.4 ✓ [2.8 — 3.8]
5: 2.6 ✓ [3.0 — 4.0]
6: 2.9 x
7: 3.4 ✓ [3.8 — 4.8]
8: 4.2 x
9: 4.7 x

M1: $0.8 + 0.4 = 1.2$
M2: $0.5 + 0.4 = 0.9$
M5: $2.6 + 0.4 = 3.0$
M6: $2.4 + 0.4 = 2.8$

1: 0.4 [0.8 — 1.8] ✓
2: 1.3 x
3: 2.2 [2.6 — 3.6] x $2.4 + 0.4 = 2.8$
4: 2.4 [2.8 — 3.8] x $2.2 + 0.4 = 2.6$
5: 2.9 x
6: 3.3 ✓ [3.8 — 4.8] $3.6 + 0.4 = 4.0$

7: 3.6 ✓ [4.0 – 5.0] 3.3 + 0.4 = 3.7

8: 4.3 X